Theorems with Chords & Tangents

Exam code: 0607

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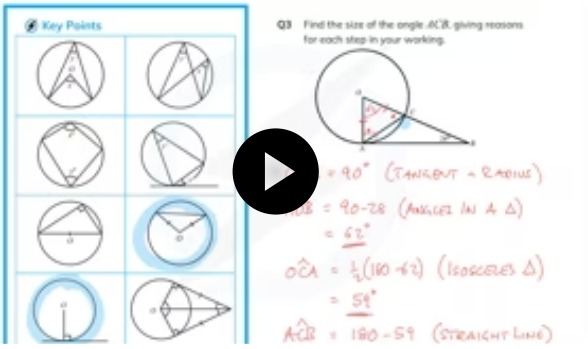

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Help me understand

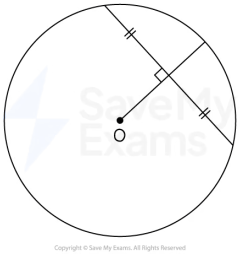
Circles & chords

What is a chord?

- A **chord** is any straight line is a circle that joins any **two points** on the **circumference**
  - Chords of **equal length** are **equidistant** (the same distance) from the centre

Circle Theorem: The perpendicular bisector of a chord passes through the centre

- If a **line through the centre** (such as a **radius or diameter**) goes through the **midpoint** of **chord**
  - it will **bisect** (cut in half) that chord at **right angles** to it



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

chord

- When explaining this theorem in an exam you can use either phrase below:
  - **A radius bisects a chord at right angles**
  - **The perpendicular bisector of a chord passes through the centre**

Examiner Tips and Tricks

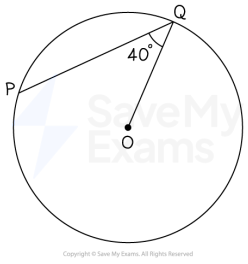
Look out for **isosceles triangles** formed by a chord and two radii.

Two angles in the triangle will be equal and there will be at least one line of symmetry.

Worked Example

The diagram below shows a circle with centre, *O*.
Two points, *P* and *Q*, lie on its circumference.
The radius of the circle is 6 cm.
Angle *OPQ* = 40°.

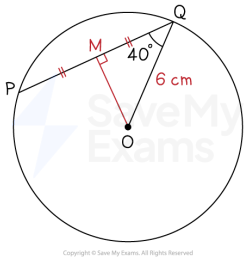
Find the length *PQ*.



Answer:

Label the radius on the diagram 6 cm

Draw a line from O to the midpoint, *M*, of the line *PQ*
The angle formed between the *OM* and *PQ* will be a right angle



$$6 \cos 40 = \frac{6}{MQ}$$

$$MQ = 4.59626...$$

Double MQ to find the length PQ

$$4.59626... \times 2 = 9.19253...$$

Round to 3 significant figures

$$PQ = 9.19 \text{ cm (3 s.f.)}$$

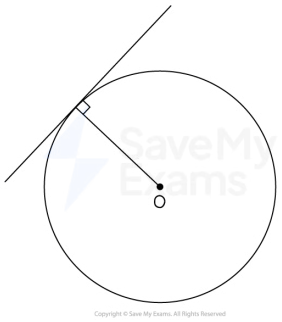
Circles & tangents

What is a tangent?

- A tangent to a circle is a straight line **outside of the circle** that touches its **circumference** at exactly **one point**

Circle Theorem: A radius and a tangent meet at right angles

- If a **radius and a tangent** meet at a point on the **circumference** of a circle, the **angle** formed between them will be **90°**
  - They are **perpendicular** to each other



- When explaining this theorem in an exam you must use the keywords:
  - **A radius and a tangent meet at right angles**

Circle Theorem: Tangents from an external point are equal in length

- **Two tangents** from the **same external point** are equal in length
- This means that a **kite** can be formed by two tangents meeting a circle
  - The kite below has a vertical line of symmetry
    - It is formed from two **congruent** triangles back-to-back
  - The kite will have **two right angles** where the tangents meet the radii

Examiner Tips and Tricks

Look for tangents in the exam and draw on the radius at right angles to see if it helps.

Worked Example

Find the value of θ in the diagram below.

Answer:

The lines *ST* and *RT* are both tangents to the circle
They meet the two radii on the circumference at the points *S* and *T*

Angle *TSO* = angle *TRO* = 90°
A radius and a tangent meet at right angles

Use vertically opposite angles to find the value of angle *RTS*

Angle *RTS* = 25°
Vertically opposite angles

Mark these angles clearly on the diagram

Angles in a quadrilateral add up to 360°
Use this to form an equation for θ

$$\theta + 90 + 90 + 25 = 360$$

Angles in a quadrilateral sum to 360°

Simplify

$$\theta + 205 = 360$$

Solve

$$\theta = 360 - 205$$

$$\theta = 155°$$

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